Journal of Experimental Psychology: General
2014, Vol. 143, No. 1, 332–339

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0096-3445/14/$12.00 DOI: 10.1037/a0031511

Learning From Social Rewards Predicts Individual Differences in Self-Reported Social Ability

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The ability to adapt face-to-face social behavior in response to an interaction's changing contingencies is an important aspect of social skill. Individual differences in social ability may depend on how well people learn from social rewards and punishments. Here we relate people's social aptitude to their ability to learn from differences in the reward values of two common social reinforcers, genuine and polite smiles. In a series of experiments, participants experienced a hidden social contingency in which they either learned to repeat actions that received genuine smile feedback and switch after polite smiles or the reverse. A condition with nonsocial feedback served as a comparison measure. Participants showed better ability to repeat actions reinforced with genuine smile feedback than with nonsocial feedback. When participants were required to switch actions following genuine smiles, performance was inhibited relative to nonsocial reinforcement. The ability to detect task contingencies and learn from social rewards predicted self-reported social ability. These novel results suggest that individual differences in reinforcement learning, and particularly in people's motivation to receive social rewards, may relate to social ability in face-to-face interactions. This finding has important implications for understanding the social difficulties that characterize disorders such as autism, depression, and schizophrenia, in which the ability to learn from rewards may be compromised.

Keywords: social rewards, smiles, learning, social cognition, social skill

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The ability to adapt behavior in order to seek social rewards and avoid punishments is an element of social competence that people begin to develop from an early age. For example, young children learn that a little mischievous behavior is amusing to their parents, and might repeat a behavior to acquire a smile. However, they also know the point at which a parent's mood will shift from amusement to anger and can alter their behavior before punishment results. Subtle, moment-to-moment changes in a parent's social cues likely provide the evidence children need to adapt behavior in this way.

In keeping with this idea, evidence suggests that people decode a range of messages from the nonverbal cues they exchange during interactions. For example, they reliably decipher emotions and intentions from facial expressions, postures and gestures (Ekman, 1993; Izard, 1992; Johnston, Miles, & Macrae, 2010). Patterns of behavior can provide similar information indicating, for example, changes in the valence or volatility of an interaction partner's state (Ames & Johar, 2009; Behrens, Hunt, Woolrich, & Rushworth, 2008; King-Casas et al., 2005). People even use cues in a deliberate attempt to shape others' behavior (Blair, 2003). Indeed, research shows that even without specific instruction, people readily repeat actions reinforced with smiles but alter their behavior in response to frowns (Kringelbach & Rolls, 2003).

This flexibility in cue use suggests that social cues serve two functions simultaneously. First, as previous research suggests, social cues are communicative, meaning that they provide receivers with information about sender's states (Ekman, 1992; Fridlund, 1991). These cues, including both their type and timing, are important information sources because they serve as a form of evidence or feedback about how well received one's recent actions have been. For example, a partner's smiles may indicate positive evaluation, agreement, and so on, whereas frowns might suggest the opposite. A range of other cues such as hesitations, head tilts, and raised eyebrows have more ambiguous meanings but may nonetheless convey understandable information to receivers.

Second, and perhaps more important, social cues should allow people to optimize their behavior to achieve the best possible outcomes. Because interactions are continual exchanges of verbal and nonverbal behavior, they require people to make frequent and rapid decisions about what to do next. The nature of a partner's reaction to one's previous behavior should enhance this decision-making process by providing information about that partner's likely reactions to possible future actions. Specifically, a receiver can use the information conveyed by a cue to make inferences about the probable outcomes of a particular behavioral strategy. Based on predictions about how a partner will view different social actions, people can choose actions with increased likelihood of reward or decreased likelihood of punishment.

Simple reinforcement-learning models offer a mechanism for predicting how moment-to-moment changes in social cues might enhance social decision-making strategies. In nonsocial settings, agents choose behaviors based on predictions about the values of possible outcomes. If an experienced outcome is better than anticipated, the expected value of the action increases, as does the

This article was published Online First January 21, 2013.

Thanks to Rachel Armer, Suzanne Cooper, Robin Kramer, and Irene Nip for helping with data collection. Economic and Social Research Council funding (RES-000-22-2262) helped to support this work.

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likelihood that the agent will repeat the action (Dayan & Balleine, 2002; Schultz, Dayan, & Montague, 1997; Seymour et al., 2004). This is known as *win-stay learning* (Berke, Breck, & Eichenbaum, 2009; Ito & Doya, 2009). *Lose-shift learning* occurs when worse-than-expected outcomes cause changes in behavioral strategy (Boureau & Dayan, 2011; Dayan, Niv, Seymour, & Daw, 2006). Thus, the value of an experienced outcome shapes expectations about the value of the associated action, which influences the likelihood that the action will be repeated (Dayan & Balleine, 2002).

In social settings, a similar process may occur on the basis of the values of social cues. Some social cues, such as smiles, are intrinsically high in reward value (Averbeck & Duchaine, 2009; Furl, Gallagher, & Averbeck, 2012), whereas others, such as frowns may be a form of punishment (Blair, 2003; Kringelbach & Rolls, 2003). Because cues differ in value, cue fluctuations may indicate subtle shifts in a partner's evaluation of one's behavior. These changes allow people to update the expected values of possible actions and subsequently choose the ones with the highest anticipated outcomes. For example, a genuine smile that becomes a polite smile might indicate that its sender's positive emotion has waned. Although such a social cue shift is not a punishment per se, a receiver might choose to view it as an indicator of change in the social climate, meaning that continuing with a particular behavioral strategy would be unwise.

Here, we provide a basic test of this idea in a study in which participants were allowed to deduce a hidden social contingency using two common social cues: genuine and polite smiles. We have previously shown that relative to polite smiles, genuine smiles are more rewarding (Shore & Heerey, 2011). If, as reinforcement-learning explanations predict, the magnitude of a reward strengthens cue–action associations, participants should learn to repeat an action reinforced with a genuine smile. Moreover, based on the finding that polite smiles are less rewarding than genuine, we anticipated that the decrease in value when a polite smile follows a genuine should facilitate a shift in behavior.

We also tested a second prediction about the ways in which the ability to learn a hidden contingency from social rewards relates to social communication skill. Recent research has begun to suggest that an individual's ability to detect contingencies and learn from rewards relates to social ability (Reeb-Sutherland, Levitt, & Fox, 2012; Solomon, Smith, Frank, Ly, & Carter, 2011). We therefore anticipated that the social reward value of genuine smiles would enhance the ability to learn a *genuine-smile–stay* contingency in participants who report better social ability on a questionnaire measure of social function, relative to those who report that their social ability is worse.

Experiment 1

In win–stay learning models, rewards facilitate repetition of the action most closely associated with their occurrence (Dayan & Balleine, 2002; Schultz et al., 1997). Because genuine smiles are rewarding relative to polite smiles (Shore & Heerey, 2011), actions that result in genuine smiles should have a greater probability of repetition than those resulting in polite smiles, even if equivalent financial gains accompany both outcomes. To test this idea, we designed a task in which participants learned a hidden contingency from genuine and polite smiles, both of which indicated correct responses and equal financial rewards. We additionally predicted that participants who learned the contingency faster would report better social–communicative ability as measured by the Autism-Spectrum Quotient (AQ) scale (Baron-Cohen, Wheelwright, Skinner, Martin, & Clubley, 2001). A number of recent studies have shown that AQ scores correlate with social perception (Baron-Cohen, Wheelwright, Hill, Raste, & Plumb, 2001; Nummenmaa, Engell, von dem Hagen, Henson, & Calder, 2012), behavior (Bayliss & Tipper, 2006; Chakrabarti et al., 2009; Hermans, van Wingen, Bos, Putman, & van Honk, 2009), and skills associated with social function, including perspective taking, empathy, and sensitivity to social contingencies (Bayliss, di Pellegrino, & Tipper, 2005; Bayliss & Tipper, 2005; Shelton, Clements-Stephens, Lam, Pak, & Murray, 2012). Together, these data suggest that the AQ is a valid measure of social–communicative skill.

Method

Participants. Forty-five undergraduate students (30 women; mean age = 20.11 years, SD = 1.57) participated in exchange for partial course credit and a small, performance-based monetary bonus (mean bonus = £3.33, SD = 0.62, or approximately US$5.39). Participants gave written informed consent before taking part. The university's ethics committee approved all research procedures (likewise for Experiment 2).

Stimuli. Face stimuli consisted of color images of actors in each of four poses (neutral/nonexpressive, polite smile, genuine smile, frown), with eye-gaze directed toward the viewer. To create these images, we invited a set of 16 actors (eight female) into the lab and asked them to produce a series of facial expressions. Actors produced eight examples of each expression, returning to neutral afterward. This meant that the resulting stimuli varied slightly in form, as do social cues in life.

We asked actors to produce polite smiles and frowns after seeing them demonstrated. To elicit spontaneous genuine smiles, we used an emotion-induction procedure in which actors recalled a series of self-relevant events that had caused them joy, amusement, or happiness and were asked to express their feelings as they would if sharing the experience with a close friend. We used a high definition digital camcorder for image capture and clipped still images from each expression at its peak and neutral baseline. We then selected the five images in each expression series (polite smiles, genuine smiles, and frowns) that were closest to a prototypical expression of that type. In this case, genuine smiles involved simultaneous action of both orbicularis oculi and zygomaticus major muscles, whereas polite smiles involved the action of the zygomaticus major muscle alone (Ekman, Davidson, & Friesen, 1990). We also selected five neutral images for each actor.

In a separate validation study, we asked 60 participants (24 male, mean age = 20.76, SD = 3.04) to view each image and state which type of expression it showed. Images were presented in two blocks (counterbalanced order), grouped by expression type. In one block, participants viewed genuine and polite smiles in pseudorandom order such that no actor appeared more than once in a row. In the other block, they viewed frowns and neutral poses. As expected, participants discriminated frowns from neutral poses with > 99% accuracy for all examples and all actors. However, this was not true of genuine and polite smiles, which are more similar in appearance. We therefore coded participants' responses

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as "hits" if they correctly identified genuine smiles and "false alarms" if they erroneously labeled a polite smile as genuine. We then used signal detection theory to compute d' as a measure of participants' ability to discriminate the smiles (Macmillan & Creelman, 2005).

Generally, participants distinguished the smiles well ($d' = 1.34$, $SD = 0.79$; for an unbiased observer, this corresponds to approximately 75% correct). We then selected the six female and six male actors for whom the average d' was closest to 75% correct. This ensured that each actor's smiles were distinguishable to a similar degree. Across the set of 12 faces, the average d' was 1.32 ($SD = 0.52$).

Procedures. To determine the degree to which participants could learn a genuine smile–stay contingency, we developed a modified version of a task designed by Kringelbach and Rolls (2003). On each trial, participants viewed two neutrally posed, same-sex faces, presented side-by-side on the computer screen. Participants were instructed to choose the face that was most likely to be correct on each trial by pressing a left or a right key. There was no reaction time cutoff for choices. Once participants made a choice, a green frame appeared around the chosen face to indicate that the computer had registered the response. After 250 ms, the chosen face either smiled to indicate a correct response or frowned to indicate an incorrect response. The feedback display remained visible for 1,000 ms, at which time the frame around the image disappeared and the face returned to neutral. The next trial began immediately after the transition from expressive display to neutral. The unselected face remained in a neutral pose throughout the trial and the same pair of faces remained on the screen for an entire block (Figure 1).

Unbeknownst to participants, the type of smile they received as feedback also predicted which face would be correct on the next trial (frowns were not predictive). If the chosen face smiled genuinely, the same face would be correct on the next trial (genuine smile–stay). If it smiled politely, the other face would be correct on the next trial. Participants were told they would win bonus money for each correct answer and were instructed to obtain as many smiles as possible. Both faces were correct on half the trials in a block in random order.

Although the same faces were continuously present throughout a block, each time an image changed, a different version of the appropriate image appeared. For example, there were five polite smiles for each face. Therefore, different images appeared when that face produced a polite smile, likewise for other displays. The exact image that the computer displayed was drawn randomly with replacement from the pool of available images. This meant that across trials, there were subtle changes in actor and cue appearance, as in real-world social environments.

All participants completed six blocks of 50 trials each. Half the blocks involved pairs of female faces, and the remainder involved male face-pairs. All 50 trials within a block showed the same face pair. The sex of the faces altered from block to block with order counterbalanced across participants. To ensure that specific face pairings did not drive the results, we paired each female face with each of the other female faces with equal frequency across participants (likewise for male faces).[1] Finally, as an indirect measure of social ability, participants completed the Autism-Spectrum Quotient (AQ) scale which measures autistic-like traits across a number of domains (Baron-Cohen, Wheelwright, Skinner, et al., 2001).

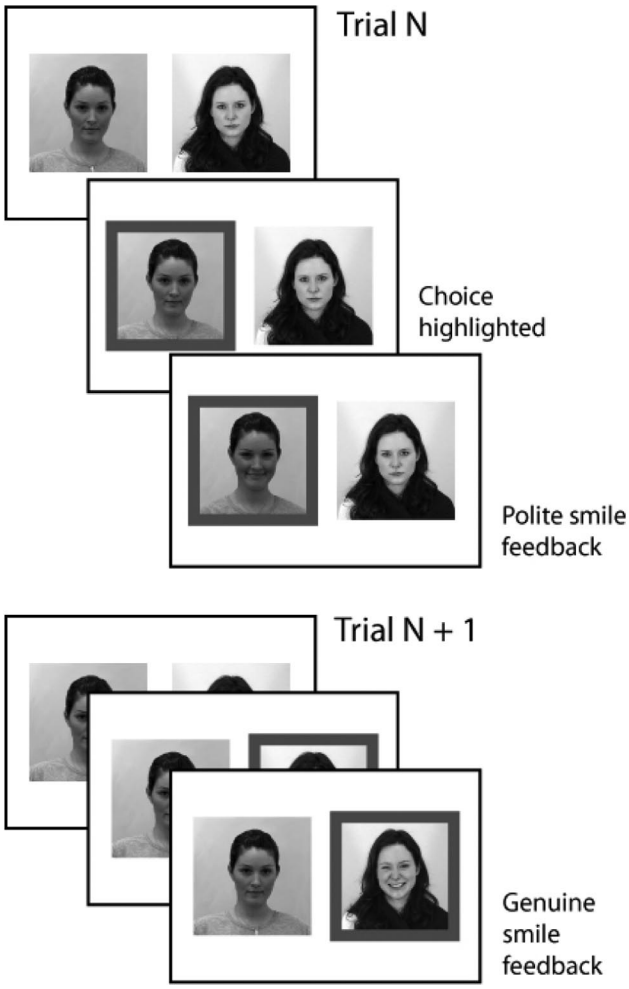

Figure 1. Trial timeline and example stimuli from Experiment 1. During the task, polite smile feedback indicated that the nonselected face would be correct on the next trial, and genuine smile feedback indicated that the chosen face would be correct again.

A coefficient alpha of .79 indicated strong internal consistency on the AQ within the sample.

Results and Discussion

Previous research suggests the presence of gender differences in both AQ and social functioning (Baron-Cohen & Wheelwright, 2004; Proverbio, Zani, & Adorni, 2008). We therefore compared male and female participants' performance. There were no gender differences in social–communicative ability based on the AQ, $t(43) = -1.60$, $p = .12$, Cohen's $d = 0.48$; overall task performance, $t(43) = 1.19$, $p = .24$, Cohen's $d = 0.37$; or genuine-smile–stay responses, $t(43) = 1.02$, $p = .31$, Cohen's $d = 0.32$. We therefore collapsed across participant gender. As Figure 2 shows, across blocks and across trials within blocks, participants

[1] The intraclass correlation coefficient between performance on the even and odd items in the task was .965, suggestive of highly consistent behavior throughout the task.

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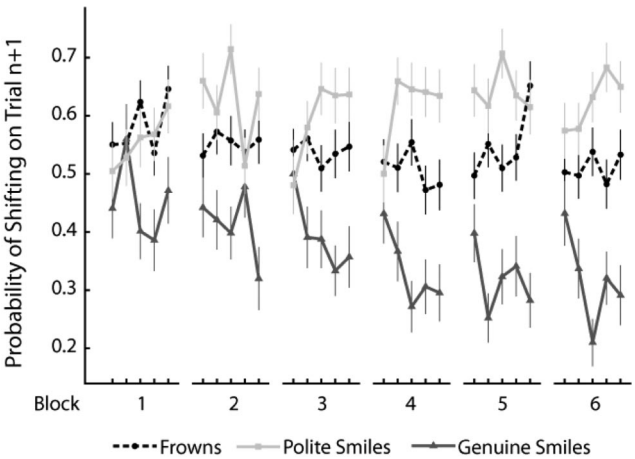

Figure 2. To determine whether participants learned to adopt a win–stay strategy after wins indicated by genuine smiles and a win–shift strategy after wins indicated by polite smiles, we split each block into five subblocks of 10 trials each and calculated the average probability of making a shift response on Trial $n + 1$ for each social cue for each subblock. Error bars show ±1 standard error of the mean.

increased their likelihood of shifting after wins indicated by polite smiles and decreased their likelihood of shift responses after genuine smiles, meaning that they deduced the task contingency. Participants who correctly articulated the learning rule ($N = 26$) achieved more correct responses, independent samples, $t(43) = 7.60$, $p < .001$, Cohen's $d = 2.46$, and produced more genuine-smile–stay responses, $t(43) = 6.00$, $p < .001$, Cohen's $d = 1.89$. Interestingly, they also reported fewer autistic-like traits (lower AQ total scores), indicative of greater social ability, $t(43) = 2.54$, $p = .02$, Cohen's $d = 0.75$.

To explore the links between social ability and task performance more specifically, we computed the correlations between participants' total AQ scores and learning on the task. Results showed that participants reporting better social–communicative ability (lower AQ scores) had better performance in the task, $r = -.40$, $p = .006$, 95% confidence interval (CI): [−.62, –.12]; and more genuine-smile–stay responses, $r = -.50$, $p = .001$, 95% CI [–.63, –.35]. This suggests that participants with better social ability were more motivated to repeat choices reinforced by genuine smiles, confirming our hypothesis that social rewards may be more motivating to people with stronger social abilities.

An alternate explanation for this finding is that participants with better social–communicative ability were simply more able to discriminate between genuine and polite smiles. To answer this question, we conducted a control study in which participants ($N = 84$, of whom 64 were women; mean age = 19.76, $SD = 2.75$) completed a smile discrimination task, followed by the AQ (a coefficient α of .79 indicated good reliability in this sample). In the task, participants saw each of the smile stimuli used in the present study (12 actors, each displaying five genuine and five polite smiles; 120 total trials) in pseudorandom order such that the same face never appeared on consecutive trials. They identified each smile as either genuine or polite with a key press. We coded correctly detected genuine smiles as "hits" and polite smiles that participants incorrectly labeled as genuine as "false alarms" and then used signal detection theory to calculate d' as an accuracy measure and criterion (Macmillan & Creelman, 2005) to determine whether participants were biased to say one type of smile over the other. Results showed that there was no correlation between social–communicative ability as measured by the AQ and either smile discrimination, $r = -.17$, $p = .13$, 95% CI [–.37, .05], or bias, $r = .07$, $p = .53$; 95% CI [–.15, .28].[2] Thus, it is not likely that the relationship between smile discrimination ability and social–communicative ability explains differences in the degree to which participants learned from social rewards.

Experiment 2

Results from Experiment 1 indicated that the reward value of a genuine smile facilitated learning of a genuine-smile–stay contingency and that the speed with which participants learned from social rewards correlated with self-reported social–communicative ability. However, because the task tested only one social contingency (genuine smile–stay), these results raise two additional questions. First, it may be the case that participants would deduce this contingency from any stimulus, meaning that these results are not specific to learning from social rewards. Second, social ability may relate to general rule learning or to the ability to deduce contingencies from moment-to-moment changes in others' social cues, rather than to a specific motivation to maximize social rewards. Experiment 2 resolves these issues by testing participants on two additional contingencies. We predicted that the genuine-smile–stay contingency will enhance learning over both a nonsocial version of the task and over a contingency in which participants must shift after genuine smile feedback.

Method

Participants. Study participants were 111 undergraduate students (63 women; mean age = 20.59 years, $SD = 3.00$) participated in exchange for partial course credit and a small performance-based monetary bonus (mean bonus = £2.48, $SD = 1.16$, or US$4.01).

Procedures. Trials in this version of the task were conducted identically to those in Experiment 1. Here, however, we randomly assigned participants to one of three contingencies. One group of participants ($N = 37$) experienced the same genuine-smile–stay contingency as in Experiment 1. For a second group of participants ($N = 36$), the opposite was true. For these participants, genuine smiles predicted that the other face would be correct on the next trial (genuine-smile–shift). Therefore, under this contingency, participants needed to inhibit responses reinforced by genuine smiles.

In order to compare the speed of learning in the social conditions to learning from nonsocial stimuli, we had a third group of participants ($N = 38$) complete the task with nonsocial stimuli. Trial procedures were identical in this version of the task; how-

[2] To ensure that the control study measured comparable levels of social ability (mean AQ = 16.21, $SD = 4.93$) as Experiments 1 (mean AQ = 14.40, $SD = 6.23$) and 2 (mean AQ = 15.27, $SD = 5.17$), we examined the differences in AQ total across the data sets. A one-way ANOVA with experiment number (Experiment 1, Experiment 2, or control) as the between-subjects variable showed that there were no differences in participants' total AQ scores across experiments, $F(2, 237) = 1.82$, $p = .17$, $\eta_p^2 = .02$.

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ever, instead of faces, participants saw pairs of empty juice glasses. As with the faces, each glass was unique and presented next to another glass from the stimulus set. On each trial, participants chose the glass on the left or the glass on the right with a key press. If they chose correctly, the glass filled with either yellow or orange juice. If they chose incorrectly, the glass filled with water. Correct responses were worth bonus money at the end of the task. In this nonsocial condition, half the participants experienced an orange juice–stay contingency and the remainder experienced an orange juice–shift contingency.

Participants completed up to four blocks of 80 trials each.[3] In order to make the task shorter for those who learned the contingencies quickly, we defined a performance-based target that exited the task at the end of a block in which participants demonstrated learning by committing no more than 10 errors in that block. A debriefing interview confirmed that 100% of the participants who reached this target were able to articulate the correct learning rule. After the learning task, participants completed the AQ (Baron-Cohen, Wheelwright, Skinner, et al., 2001) as a measure of individual differences in social ability (coefficient α within the sample was .77).

Results and Discussion

Dependent measures in the learning task included the proportion of trials on which participants responded correctly and the number of trials required for participants to reach a *learning criterion*, defined as 10 consecutive correct responses, with at least three shifts in the sequence. As in Experiment 1, there were no gender differences in (a) task performance: trials to criterion: $F(1, 105) = 2.04$, $p = .16$, $\eta_p^2 = .02$; (b) proportion correct: $F(1, 105) = 1.87$, $p = .17$, $\eta_p^2 = .02$; or (c) social–communicative ability, $F(1, 105) < .01$, $p = .99$, $\eta_p^2 < .001$. We therefore collapsed across participant gender. A one-way analysis of variance (ANOVA) demonstrated significant differences between task conditions in the number of trials participants required to reach criterion, $F(2, 108) = 14.22$, $p < .001$, $\eta_p^2 = .21$ (Figure 3A). Post hoc comparisons indicated that participants in the genuine-smile–stay condition learned the task significantly faster than did those in either of the other conditions: ps $< .01$, Cohen's d values $> .82$; Bonferroni-corrected to control Type I error (likewise for comparisons reported later). Participants in the nonsocial and genuine smile shift conditions did not significantly differ ($p = .19$, Cohen's $d = .39$).

We confirmed these results by examining the proportion of trials on which participants chose correctly (Figure 3B shows performance across task blocks). Results showed significant group differences, $F(2, 108) = 12.95$, $p < .001$, $\eta_p^2 = .19$. Participants in the genuine-smile–stay condition had more correct answers than did those in other conditions, ps $< .03$, Cohen's d values $> .60$. In addition, participants in the genuine-smile–shift condition performed more poorly than did those in the nonsocial condition, $p = .05$, Cohen's $d = .58$. Together, these results suggest that the greater social value associated with genuine relative to polite smiles enhanced the likelihood of stay responses, meaning that the genuine-smile–stay contingency was easier to detect than the basic nonsocial learning contingency and that the genuine-smile–shift condition inhibited learning.

To relate self-reported social ability with learning, we correlated participants' total AQ scores with trials to criterion and proportion correct within each group (Figure 4). To recall our prediction, if general rule-based or contingency learning underpins social ability, the correlations between task performance and social–communicative ability should be significant across the three groups. If social ability relates specifically to the ability to learn from subtle changes in social cues, the correlations should be significant in the social feedback groups only. Finally, if social ability relates to the degree to which social rewards drive learning, only the group who experienced the genuine-smile–stay contingency should show a significant correlation between social ability and performance. As Table 1 shows, only participants in the genuine-smile–stay condition showed strong and significant correlations between self-reported social–communicative ability and both learning measures (trials-to-criterion and proportion correct). These results suggest that the ability to learn from facial expression-based social rewards may be an important aspect of individual variability in social and communicative skill.

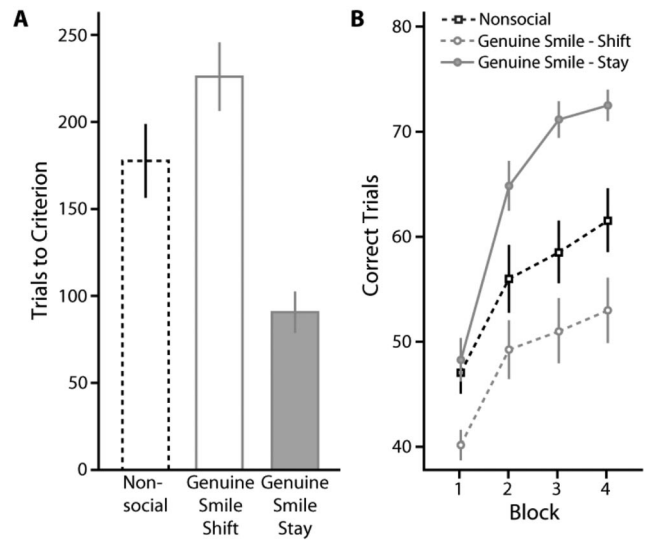

Figure 3. Learning across reward contingencies. (a) The number of trials that participants were required to reach a performance-based criterion. (b) Learning over task blocks (for participants who completed fewer than four blocks because they learned faster, scores on the final block were carried forward). Error bars show ±1 standard error of the mean.

To test this idea statistically, we used Fisher's transformation to convert the correlations to z scores and examined the group differences. As our prediction was directional, we report one-tailed tests. For our trials-to-criterion measure, participants who experienced the genuine-smile–stay contingency showed a stronger correlation between performance and social–communicative skill than did those experiencing the nonsocial task, $z = 1.71$, $p = .04$, and tended to do so with respect to those in the genuine-smile–shift condition, $z = 1.52$, $p = .06$. Similarly, participants in the genuine-smile–stay condition showed a stronger correlation between social–communicative skill and proportion correct than did those in the genuine-smile–shift condition, $z = 1.77$, $p = .04$. This relationship was also stronger in genuine smile-stay condition than

[3] The intraclass correlation for the even versus odd items in the task was .968, suggesting reliable task performance.

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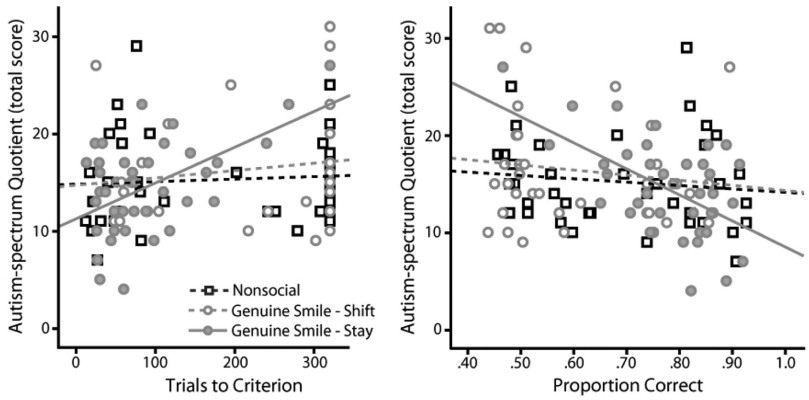

Figure 4. Correlations between social–communicative ability and task performance: (a) trials to criterion (for participants who did not learn the contingency, trials to criterion was calculated as 320 trials, the total number of trials completed); (b) proportion correct. Note that higher scores on the Autism-Spectrum Quotient (AQ) suggest worse social–communicative ability.

in the nonsocial condition, z = 1.62, p = .05. Together, these results suggest that social ability predicts the degree to which participants use social feedback and learn from social rewards.

General Discussion

The degree of flexibility with which people respond to social cues suggests that those cues simultaneously allow them to evaluate the outcomes of recent actions and to predict the quality of a social partner's likely responses to possible future actions. Based on these predictions, people can adjust their social behavior to increase the likelihood of receiving a reward and reduce the likelihood of punishment. The present results suggest that subtle shifts in the relative values of social cues can serve this function. Specifically, when a current cue is more valuable than one seen previously, people show a bias to repeat the associated action. When it is less valuable than a previous cue, they tend to shift actions. In accord with reinforcement-learning models (Behrens et al., 2008; Dayan & Balleine, 2002; Seymour et al., 2004), participants showed enhanced learning for contingencies consistent with these biases and impaired learning for contingencies that opposed these biases, relative to nonsocial cues.

The data from Experiment 1 show that the ability to detect a hidden contingency from social feedback correlated with self-reported social-communicative ability as measured by participants' responses on the social and communication skills subscales of the AQ Scale. Indeed, participants who were able to articulate the contingency reported fewer autistic-like traits. More specifically, participants' bias to repeat actions reinforced with genuine smiles correlated with their social-communicative ability, suggesting that the ability to use social reinforcement to guide behavior may be an important aspect of social skill.

Experiment 2 replicates these results and shows more specifically that social reinforcement seems to be important in driving these effects. Findings are consistent with research showing that genuine smiles carry social reinforcement value (Averbeck & Duchaine, 2009; Shore & Heerey, 2011) and therefore enhance win–stay learning. Both genuine and polite smiles indicated correct responses and, in financial terms, were worth the same amount. Therefore, the fact that participants quickly detected this complex genuine-smile–stay/polite-smile–shift contingency suggests that they did indeed track subtle differences in social value from one trial to the next and used this information to inform subsequent social decisions. The fact that participants in the genuine-smile–shift condition showed inhibited performance strengthens this idea. The presence of highly rewarding genuine-smile feedback reduced participants' ability to shift their responses, even though both smiles were equally financially valuable. In addition, the correlations between social–communicative

Table 1
Correlations Between Self-Reported Social–Communicative Ability and Task Performance in Experiment 2

	Trials to criterion				Proportion correct			
			95% confidence interval				95% confidence interval	
Variable	r	p	Lower bound	Upper bound	r	p	Lower bound	Upper bound
Nonsocial (n = 38)	.11	.50	−.22	.42	−.17	.29	−.46	.16
Genuine smile–shift (n = 36)	.15	.27	−.19	.46	−.13	.44	−.44	.21
Genuine smile–stay (n = 37)	.48	.002	.185	.70	−.51	.001	−.72	−.22

Note. Social–communicative ability is measured by participants' total scores on the Autism-Spectrum Quotient Scale (Baron-Cohen, Wheelwright, Skinner, et al., 2001). Higher scores on the scale suggest worse social–communicative ability. The ps reported in the table are uncorrected. The Bonferroni-corrected critical p = .008 (α = .05, n = 6).

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ability and task performance were strongest in the group experiencing the genuine-smile–stay contingency, suggesting that the ability to learn from social rewards is an important contributor to individual differences in social ability.

At a broader level, the ability to integrate the values of a partner's visible social cues over time is an important skill, especially if it informs one's future social strategy. The present results suggest this is an important use of social information. Subtle, trial-to-trial changes in the cues each face provided predicted the next trial outcome, as they might predict a partner's reaction to one's next behavior. Participants relied on these shifts when choosing their next response. A similar process of evaluating and integrating social feedback is likely to occur in real-world social interactions as well. The presence of such a process would allow people to anticipate the outcomes of possible actions and should therefore enhance social decision making. Although social predictions do not always turn out to be accurate, they nonetheless indicate the likelihood of receiving social rewards and/or punishments, thereby allowing people to optimize their social behavior to achieve the best possible outcomes.

An important limitation of this work is the fact that although genuine and polite smiles are common social cues (Johnston et al., 2010), people produce a much larger range of social behaviors. Moreover, smiles (especially genuine ones) may be particularly iconic in terms of both their form and meaning (Ekman, 1992). Thus, the cues present in natural interactions have greater variety in appearance, as well as meanings that are more ambiguous. Nonetheless, the fact that genuine smiles so readily facilitated "stay" behavior suggests that participants were predisposed to evaluate smiles in terms of their relative values as the cues the faces presented changed from trial to trial. Although this idea requires evaluation under more naturalistic conditions, these laboratory findings provide evidence that reinforcement-learning models may be a viable way of understanding individual differences in people's ability to regulate social behavior using their partners' nonverbal cues in the real world.

Finally, we note that our measure of social-communicative ability relies on people's ability to make accurate self-reports of their social skill. Although research suggests that the AQ is a valid measure of social and communicative symptoms of autism and, more broadly, the degree to which people express autism-spectrum traits (Baron-Cohen, Wheelwright, Skinner, et al., 2001), it is prone to the same issues as self-report measures generally are. This measure also equates social ability with autism-spectrum traits. Although the questionnaire does appear to measure important aspects of real-world social ability (e.g., reciprocity, conversational skill), we acknowledge that there may be different interpretations of this result.

Conclusions

Participants in the present research used subtle differences in the relative values of genuine and polite smiles to extract a hidden social contingency. As reinforcement-learning models predict, these data suggest that the reward values of genuine, relative to polite, smiles enhanced win–stay learning and the degree to which they did so correlated with social–communicative ability. These results hint that the deficits in social interaction in conditions such as autism may stem from reinforcement learning difficulties, as recent research suggests (Solomon et al., 2011). Moreover, if deficits in the ability to learn from reinforcement do indeed characterize disorders such as depression (Chentsova-Dutton & Hanley, 2010; Der-Avakian & Markou, 2012), these results suggest a mechanism underpinning the acquired social difficulties associated with this illness. Taken together, these findings suggest that reinforcement learning both supports the moment-to-moment decisions that social actors must make and is an important aspect of social ability.

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Received May 20, 2012
Revision received December 10, 2012
Accepted December 11, 2012 ■

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